National University of Computer and Emerging Sciences

FAST School of Computing   Spring-2025   Islamabad Campus

Artificial Intelligence — Sample Paper 1   [Total Marks: 40]   Time: 75 Minutes

Question 1 [15 Marks] — Constraint Satisfaction Problem

Five tasks T1–T5 must be assigned start hours from the domain {8, 9, 10, 11} (24-hour clock, each task takes exactly 1 hour). The variables are T1, T2, T3, T4, T5. The constraints are:

C1: $T1 + 1 < T3$

C2: $T2 + 2 < T5$

C3: $T3 \neq T4$

C4: $T4 + 1 < T5$

C5: $T1 \neq T2$

(a) Identify all **unary constraints** in the list above. Justify your answer. [2]

(b) Draw the **constraint graph**. Label every node (variable) and every edge (constraint). [3]

(c) Enforce **arc consistency (AC-3)** on the original domains. Show the queue operations and the final reduced domains for each variable. [5]

(d) Using the reduced domains from (c), apply **backtracking search** with the variable order T1, T2, T3, T4, T5 and **least-constraining-value** ordering. Show the complete backtracking table and state the first valid full assignment found (or state failure). [5]

Question 2 [15 Marks] — Genetic Algorithm

A genetic algorithm uses chromosomes of the form x = abcdefgh (length 8). Each gene is a digit 0–9. The fitness function is:

$$f(x) = (a + b)^2 - (c + d) + (e - f)^2 - (g + h)$$

The initial population is:

x1 = 5 3 7 1 4 2 8 6

x2 = 9 0 2 5 3 1 7 4

x3 = 2 6 4 3 9 0 5 1

x4 = 7 1 8 2 0 6 3 9

(a) Evaluate the fitness of each individual. Rank them fittest to least fit. [3]

(b) Cross the **two fittest** individuals using **one-point crossover at the middle** (between positions 4 and 5). Show the two offspring. [2]

(c) Cross the **2nd and 3rd fittest** using **two-point crossover at positions b and f** (i.e., after index 1 and after index 5). Show the two offspring. [2]

(d) Cross the **1st and 3rd fittest** using **uniform crossover** with the mask A B A B B A A B. Show the two offspring. [2]

(e) The new population consists of the six offspring from (b), (c), and (d). Compute the **average fitness** of the new population. Has it improved vs the original? Show all working. [4]

(f) Find the **chromosome with maximum possible fitness** (genes 0–9 only). State the chromosome and its fitness value. [2]

Question 3 [10 Marks] — Minimax & Alpha–Beta Pruning

The game tree below has MAX at the root. Leaf values are given. The branching factor is 3 at every level. Levels alternate MAX $\rightarrow$ MIN $\rightarrow$ MAX $\rightarrow$ MIN (4 levels of non-leaf nodes).

Subtree	Leaf values (left to right)
Branch A – left sub	12, 3, 8
Branch A – mid sub	5, 17, 2
Branch A – right sub	9, 6, 14
Branch B – left sub	7, 1, 11
Branch B – mid sub	4, 20, 3
Branch B – right sub	13, 5, 8
Branch C – left sub	6, 2, 15
Branch C – mid sub	10, 7, 4
Branch C – right sub	3, 18, 1

(a) Using **Minimax only** (no α–β), fill in the values of **all** internal nodes. State the root value and the optimal first move. [4]

(b) Apply α**–**β **pruning** (left-to-right evaluation). Indicate which branches are pruned and in what order leaf nodes are evaluated. State the final α and β at the root after processing. [4]

(c) How many leaf nodes does α–β evaluate vs Minimax (with perfect ordering)? What is the theoretical minimum? [2]

Artificial Intelligence — Sample Paper 2 [Total Marks: 40] Time: 75 Minutes

Question 1 [15 Marks] — Constraint Satisfaction Problem

Ahmed is scheduling 4 meetings (M1, M2, M3, M4) into 4 one-hour time slots: {9, 10, 11, 12}. Each meeting must be assigned exactly one slot. The constraints are:

C1: M1 must happen before M3 (M1 < M3)

C2: M2 must happen before M4 (M2 < M4)

C3: M1 and M2 cannot be in the same slot (M1 ≠ M2)

C4: M3 + 1 < M4

C5: M2 + 2 ≤ M3

(a) List which constraints are **binary** and which (if any) are **unary**. [2]

(b) Show the **AC-3 arc consistency** process step by step. Write the initial queue, show each arc processed, and state the final domains. [6]

(c) Using the reduced domains, run **backtracking** with variable order M1, M2, M3, M4 and **MRV heuristic** for tie-breaking. Show the full backtracking table and the complete solution or report failure. [7]

Question 2 [15 Marks] — Genetic Algorithm (TSP)

Solve a 5-city TSP using a GA. Cities are C1–C5. Distance matrix (symmetric):

	C1	C2	C3	C4	C5
C1	0	10	15	20	25
C2	10	0	12	18	14
C3	15	12	0	9	22
C4	20	18	9	0	11
C5	25	14	22	11	0

Parents:

Parent 1: [C1, C3, C5, C2, C4]

Parent 2: [C4, C2, C1, C5, C3]

(a) Compute the **total tour distance** (fitness) for both parents. The tour is circular (return to start). [4]

(b) Apply **Order Crossover (OX)** with crossover segment from position 2 to 4 (1-indexed). Show clearly how duplicates are resolved for both children. [4]

(c) Apply **swap mutation** on Child 1 by exchanging positions 1 and 4. Compute the mutated child's tour distance. [3]

(d) Compute the fitness of Child 2 (no mutation). Has the crossover improved the population's **best** individual? [2]

(e) Explain briefly why crossover alone cannot always reach the optimal tour in TSP. [2]

Question 3 [10 Marks] — Alpha–Beta Pruning

A two-player zero-sum game has the following game tree (MAX at root, depth = 3 half-moves). All nodes at depth 1 are MIN nodes, depth 2 are MAX nodes, depth 3 are leaf nodes.

Node	Children (leaf values or node label)
Root (MAX)	A, B, C
A (MIN)	A1=14, A2=3, A3=7
B (MIN)	B1=2, B2=19, B3=11
C (MIN)	C1=5, C2=8, C3=16

(a) Apply **Minimax**. Write the value of each node (A, B, C and root). What is the best move for MAX from the root? [3]

(b) Apply α**–**β **pruning** (left to right). For each node, write the α and β values as they are updated. Circle any pruned leaf nodes. [5]

(c) What are the final α and β at the root node after α–β completes? How many leaves were pruned? [2]

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Artificial Intelligence — Sample Paper 3   [Total Marks: 40]   Time: 75 Minutes

Question 1 [15 Marks] — Constraint Satisfaction Problem

Six employees (E1–E6) must each be assigned to exactly one of four project teams: {Alpha, Beta, Gamma, Delta}. Multiple employees can be on the same team. Team assignment is the domain of each variable. The constraints are:

C1: E1 ≠ E2 (cannot be on the same team)

C2: E2 ≠ E3

C3: E3 ≠ E4

C4: E1 ≠ E4

C5: E5 ≠ E1 and E5 ≠ E3

C6: E6 ≠ E2 and E6 ≠ E4

C7: E5 ≠ E6

C8 (Unary): E1 must be on Alpha or Beta only

C9 (Unary): E4 must be on Beta or Gamma only

(a) Enforce all **unary constraints** first. Write the reduced domains for all variables. [2]

(b) Draw the **constraint graph**. Clearly label all nodes (E1–E6) and edges (binary constraints only). [3]

(c) Run **AC-3** starting from the domains obtained in (a). Show the queue and all domain reductions step by step. Write the final domains. [5]

(d) Assign E1 = Alpha. Perform **forward checking** — show which values are eliminated from the domains of all neighbours of E1. Then assign E2 = Gamma and perform forward checking again. [5]

Question 2 [15 Marks] — Genetic Algorithm (N-Queens)

An 8-Queens GA uses chromosomes of length 8 where chromosome[i] is the row (1–8) of the queen in column i+1. Fitness = number of **non-attacking pairs** (max = 28). Two queens attack if they share the same row OR are on the same diagonal (|col_i − col_j| == |row_i − row_j|).

Initial population:

P1: [3, 6, 2, 7, 1, 4, 8, 5]

P2: [5, 2, 8, 1, 4, 7, 3, 6]

P3: [1, 7, 4, 6, 8, 2, 5, 3]

P4: [6, 3, 5, 8, 1, 4, 2, 7]

(a) Compute the fitness f(P) for **each** of the four chromosomes. Show all attacking pairs found. Rank from fittest to least fit. [6]

(b) Perform **single-point crossover** between P1 and P2 at split point after index 3 (columns 1–4 from parent 1, columns 5–8 from parent 2). Apply the **repair function**: replace duplicates in the second half with missing values in ascending order. Show both offspring. [4]

(c) Apply **two-point crossover** between P3 and P4 at split points after index 2 and after index 5. Apply repair. Show both offspring. [3]

(d) With a 0% mutation rate, is it possible for the GA to produce the chromosome [1, 5, 8, 6, 3, 7, 2, 4] (a known optimal solution) starting from the given population? Justify your answer by examining the gene values in each column. [2]

Question 3 [10 Marks] — Minimax with Evaluation Function

A two-player board game uses the following **static evaluation function** (positive favours MAX):

- +5 for each MAX piece still on the board
- –5 for each MIN piece still on the board
- +3 if MAX controls the centre (MAX piece at position C3 or C4)
- –3 if MIN controls the centre
- +8 if MAX threatens a WIN (can win in one move)
- –8 if MIN threatens a WIN

The game tree (MAX at root, depth-2 search, branching factor 3) produces the following board states at the leaves. Evaluate each leaf using the function above, then answer the questions.

Leaf	MAX pieces	MIN pieces	MAX centre?	MIN centre?	MAX threatens?	MIN threatens?
L1	4	3	Yes	No	No	No
L2	3	4	No	Yes	No	Yes
L3	5	2	Yes	No	Yes	No
L4	3	3	No	No	No	No
L5	2	4	No	Yes	No	Yes
L6	4	3	Yes	No	No	No
L7	4	2	No	No	Yes	No
L8	3	4	No	No	No	No
L9	5	1	Yes	No	Yes	No

Tree structure: Root → [Node A, Node B, Node C]. A → [L1, L2, L3]; B → [L4, L5, L6]; C → [L7, L8, L9]. Node A, B, C are MIN nodes.

(a) Compute the **static evaluation score** for each of the 9 leaf nodes. [4]

(b) Apply **Minimax**. Write the value of each of A, B, C, and the root. What is MAX's best move? [3]

(c) Apply α**–**β **pruning** (left to right). Mark any pruned leaf(ves). State final α and β at root. [3]